NEET UG 2027: Master Practical Handbook

Physics Unit 20: 18 Prescribed Experimental Lab Skills

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1 Vernier Calipers & Screw Gauge

Vernier Calipers: Least Count, Zero Error & Measurement

1. Least Count (Vernier Constant, VC): Smallest distance measurable by the instrument. If n Vernier Scale Divisions (VSD) equal m Main Scale Divisions (MSD):

$$\text{LC} = \mathbf{1}\,\text{MSD} - \mathbf{1}\,\text{VSD} = \left(\mathbf{1} - \frac{\mathbf{m}}{\mathbf{n}}\right)\text{MSD} \tag{1}$$

Standard Metric Caliper: $10\,\text{VSD} = 9\,\text{MSD}$ with $1\,\text{MSD} = 1\,\text{mm} = 0.1\,\text{cm}$:

$$\text{LC} = 1\,\text{mm} - \frac{9}{10}\,\text{mm} = \mathbf{0.1}\,\text{mm} = \mathbf{0.01}\,\text{cm} \tag{2}$$

2. True Reading Formulation & Zero Error Corrections:

$$\text{Observed Reading} = \text{MSR} + (\text{VSR} \times \text{LC}) \qquad \text{True Reading} = \text{Observed Reading} - (\text{Zero Error}) \tag{3}$$

- **Positive Zero Error ($+e$):** Vernier zero lies to the **right of main scale zero** when jaws are closed.

$$e = +(n \times \text{LC}) \implies \text{True Reading} = \text{Observed} - e \quad \text{(Must be subtracted)} \tag{4}$$

- **Negative Zero Error ($-e$):** Vernier zero lies to the **left of main scale zero** when jaws are closed.

$$e = -(N - n) \times \text{LC} \implies \text{True Reading} = \text{Observed} - (-e) = \text{Observed} + |e| \quad \text{(Must be added)} \tag{5}$$

Screw Gauge (Micrometer): Pitch & Backlash Error

1. Pitch & Least Count (LC):

$$\text{Pitch} = \frac{\text{Distance moved on linear pitch scale}}{\text{Number of complete rotations of circular cap}} \qquad \text{(Usually 1 mm or 0.5 mm)} \tag{6}$$

$$\text{LC} = \frac{\text{Pitch}}{\text{Total number of circular scale divisions } (N)} = \frac{1\,\text{mm}}{100} = \mathbf{0.01}\,\text{mm} = \mathbf{0.001}\,\text{cm} = \mathbf{10}\,\mu\text{m} \tag{7}$$

2. True Thickness / Diameter Calculation:

$$\text{True Reading} = \text{PSR} + (\text{CSR} \times \text{LC}) - (\text{Zero Error}) \tag{8}$$

- **Positive Zero Error:** Circular zero is **below the reference line** when studs touch $\implies e = +(n \times \text{LC})$.
- **Negative Zero Error:** Circular zero is **above the reference line** when studs touch $\implies e = -(N - n) \times \text{LC}$.
- **Backlash Error:** Due to loose thread fitting between screw and nut. Avoided by turning screw **strictly in one direction only**.

2 Mechanics & Hydrodynamics Lab Experiments

Simple Pendulum, Metre Scale & Elasticity (Searle's Apparatus)

1. Simple Pendulum (L vs T^2 Graph): Time period: $T = 2\pi\sqrt{\frac{L}{g}} \implies T^2 = \left(\frac{4\pi^2}{g}\right) L$.

- Graph of L versus T^2 is a **straight line passing through the origin**.

$$\mathbf{g = 4\pi^2\left(\frac{L}{T^2}\right) = \frac{4\pi^2}{\text{Slope of } T^2 \text{ vs } L \text{ graph}}} \tag{9}$$

- **Effective Length:** $L = l + h + r$ (l = wire length, h = hook length, r = bob radius).
- Angular amplitude must be small ($\theta \leq 4°$) so that $\sin\theta \approx \theta$.

2. Metre Scale (Principle of Moments / Determination of Mass): A balanced uniform metre scale obeys rotational equilibrium: $\sum \tau_{\text{ext}} = 0$.

$$m_1 g(x_{\text{knife}} - x_1) = Mg(x_{\text{knife}} - x_{\text{cg}}) + m_2 g(x_2 - x_{\text{knife}}) \tag{10}$$

Where M is mass of metre scale, x_{cg} is center of gravity (usually 50 cm mark for uniform scale).

3. Young's Modulus of Elasticity (Searle's Apparatus): Two identical wires (reference and experimental) avoid thermal expansion errors:

$$\mathbf{Y = \frac{Stress}{Strain} = \frac{MgL}{\pi r^2 l}} \implies \text{Elongation } l = \left(\frac{gL}{\pi r^2 Y}\right) M \tag{11}$$

Graph of elongation l versus suspended mass M is a straight line. $Y = \frac{gL}{\pi r^2 (\text{Slope})}$.

Surface Tension, Viscosity & Speed of Sound in Air

1. Surface Tension of Water by Capillary Rise Method:

$$\mathbf{T = \frac{r(h + r/3)\rho g}{2\cos\theta} \approx \frac{rh\rho g}{2}} \quad \text{(For pure water in clean glass, contact angle } \theta \approx 0°, \cos\theta = 1) \tag{12}$$

Meniscus correction adds $r/3$ to measured height h. Internal radius r measured by travelling microscope.

2. Viscosity by Terminal Velocity (Stokes' Law): Spherical lead/steel ball falling through viscous medium (glycerine/castor oil) reaches constant terminal velocity v_t:

$$F_{\text{viscous}} = 6\pi\eta r v_t = \frac{4}{3}\pi r^3(\rho - \sigma)g \implies \mathbf{\eta = \frac{2r^2(\rho - \sigma)g}{9v_t}} \tag{13}$$

Where ρ is density of sphere, σ is density of liquid. Graph of v_t versus r^2 is a straight line through the origin.

3. Speed of Sound in Air using Resonance Tube (End Correction): Water level provides closed end. For first two resonant air columns of lengths l_1 and l_2 with tuning fork of frequency f:

$$l_1 + e = \frac{\lambda}{4} \qquad l_2 + e = \frac{3\lambda}{4} \implies l_2 - l_1 = \frac{\lambda}{2} \tag{14}$$

$$\mathbf{v = f\lambda = 2f(l_2 - l_1)} \qquad \text{End Correction } \mathbf{e = \frac{l_2 - 3l_1}{2} \approx 0.6\,r} \ (r = \text{internal tube radius}) \tag{15}$$

Calorimetry & Newton's Law of Cooling

1. Specific Heat Capacity of Solid by Method of Mixtures: Heat lost by hot solid = Heat gained by water and calorimeter (at thermal equilibrium):

$$m_s s_s(T_s - T) = (m_w s_w + m_c s_c)(T - T_0) \implies \mathbf{s_s = \frac{(m_w s_w + m_c s_c)(T - T_0)}{m_s(T_s - T)}} \tag{16}$$

Where $m_c s_c = W$ is water equivalent of calorimeter. **2. Cooling Curve (Newton's Law of Cooling):**

$$\frac{dT}{dt} = -K(T - T_0) \implies \mathbf{\ln(T - T_0) = -Kt + C} \tag{17}$$

Graph of $\ln(T - T_0)$ versus time t is a straight line with negative slope $(-K)$. Graph of temperature T vs time t is an exponential decay curve flattening at ambient temperature T_0.

3 Electricity & Magnetism Lab Experiments

Ohm's Law & Metre Bridge (Wheatstone Bridge)

1. Verification of Ohm's Law ($V - I$ Graph):

$$V = IR \implies R = \text{Slope of } V \text{ vs } I \text{ graph} \qquad \text{Resistivity } \rho = \frac{\mathbf{RA}}{\mathbf{L}} = \frac{\pi \mathbf{r^2 R}}{\mathbf{L}} \tag{18}$$

2. Metre Bridge (Null Deflection Method): Uniform high-resistance manganin/constantan wire of length 100 cm:

$$\frac{P}{Q} = \frac{R}{S} \implies \frac{l_1}{100 - l_1} = \frac{R}{X} \implies \mathbf{X = R\left(\frac{100 - l_1}{l_1}\right)} \tag{19}$$

End Errors & Precautions:

- Resistance of thick copper strips at ends introduces **End Errors** (α, β); corrected by swapping R and X.
- Maximum sensitivity occurs when balance point is near the center: $\mathbf{l_1 \approx 50}$ cm $(40\,\text{cm} \le l_1 \le 60\,\text{cm})$.

Galvanometer Resistance, Figure of Merit & Conversions

1. Half-Deflection Method for Galvanometer Resistance (G):

$$\text{Initial deflection } \theta \text{ with series } R: \quad I_1 = \frac{E}{R + G} = k\theta \tag{20}$$

$$\text{Shunting galvanometer with } S \text{ reduces deflection to } \frac{\theta}{2}: \quad \mathbf{G = \frac{R \cdot S}{R - S}} \tag{21}$$

2. Figure of Merit (k): Current required to produce unit division deflection (1 div):

$$\mathbf{k = \frac{I}{\theta} = \frac{E}{(R + G)\theta}} \implies \text{Full Scale Current } I_g = n \cdot k \quad (n = \text{total scale divisions}) \tag{22}$$

3. Conversion of Galvanometer into Ammeter and Voltmeter:

- **Ammeter (Range $0 \to I$):** Connect small resistance (**Shunt** S) in **parallel** with galvanometer:

$$(I - I_g)S = I_g G \implies \mathbf{S = \frac{I_g G}{I - I_g}} \approx \frac{I_g G}{I} \tag{23}$$

  Resistance of ideal ammeter = 0. Equivalent ammeter resistance: $R_A = \frac{GS}{G+S} < S$.

- **Voltmeter (Range $0 \to V$):** Connect high resistance (R_V) in **series** with galvanometer:

$$V = I_g(G + R_V) \implies \mathbf{R_V = \frac{V}{I_g} - G} \tag{24}$$

  Resistance of ideal voltmeter = ∞. Equivalent voltmeter resistance: $R_{\text{eff}} = G + R_V$.

4 Optics & Semiconductor Lab Experiments

Focal Length of Mirror/Lens, Prism Deviation & Glass Slab

1. Focal Length by $u-v$ Method (Convex Lens & Concave Mirror):

$$\text{Convex Lens: } \frac{1}{f}=\frac{1}{v}-\frac{1}{u} \implies \text{Graph of } \frac{1}{v} \text{ vs } \frac{1}{u} \text{ has slope } -1 \text{ and intercepts } \frac{1}{f} \tag{25}$$

$$\text{Concave Mirror: } \frac{1}{f}=\frac{1}{v}+\frac{1}{u} \implies \text{At } u=2f,\ v=2f \quad \text{(Object at center of curvature)} \tag{26}$$

2. Refractive Index of Prism ($i-\delta$ Curve): Graph of angle of incidence i versus deviation δ is U-shaped with a unique minimum deviation δ_m:

$$\mu=\frac{\sin\left(\frac{\mathbf{A}+\delta_\mathbf{m}}{\mathbf{2}}\right)}{\sin\left(\frac{\mathbf{A}}{\mathbf{2}}\right)} \qquad \text{(At minimum deviation: } i_1=i_2,\ r_1=r_2=A/2,\ \text{ray inside is parallel to base)} \tag{27}$$

3. Refractive Index of Glass Slab using Travelling Microscope:

$$\mu=\frac{\text{Real Thickness}}{\text{Apparent Thickness}}=\frac{\mathbf{R_3-R_1}}{\mathbf{R_3-R_2}} \tag{28}$$

Where R_1 = reading on ink mark on paper, R_2 = reading through glass slab, R_3 = reading of lycopodium powder on top surface.

Semiconductor Diode Characteristics (Forward & Reverse Bias)

- **Forward Bias Characteristics:** p-side connected to positive, n-side to negative terminal.
  - Depletion layer width decreases; barrier potential opposes applied voltage.
  - **Knee / Cut-in Voltage (V_k):** Current is negligible until threshold voltage is reached (**0.7** V for Silicon, **0.3** V for Germanium), beyond which current rises exponentially.
  - **Dynamic Forward Resistance:** $\mathbf{r_f}=\frac{\mathbf{\Delta V_f}}{\mathbf{\Delta I_f}}$ (typically $10-100\,\Omega$).
- **Reverse Bias Characteristics:** p-side connected to negative, n-side to positive terminal.
  - Depletion layer width widens; tiny constant **Reverse Saturation Current** (I_0) flows due to minority carriers (microamperes μA for Ge, nanoamperes nA for Si).
  - **Reverse Breakdown Voltage (V_{br}):** Sudden sharp increase in current due to avalanche/Zener breakdown.
  - **Dynamic Reverse Resistance:** $\mathbf{r_r}=\frac{\mathbf{\Delta V_r}}{\mathbf{\Delta I_r}}$ (typically $>1\,\text{M}\Omega$).